Similar questions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 **In this question you must show detailed reasoning.** **[4]**

Solve the equation $4x^4 - 13x^2 + 9 = 0$.

2

(a) Solve $x - 5\sqrt{x} + 6 = 0$. **[3]**

(b) Solve $2^{2x+1} - 9 \times 2^x + 4 = 0$. **[5]**

Question 1

Let $u = x^2$: $4u^2 - 13u + 9 = 0 \Rightarrow (4u - 9)(u - 1) = 0$, so $u = \frac{9}{4}$ or $u = 1$. Hence $x = \pm\frac{3}{2}$ or $x = \pm 1$

Question 2

(a) Let $u = \sqrt{x}$: $u^2 - 5u + 6 = 0 \Rightarrow (u - 2)(u - 3) = 0$, so $u = 2$ or 3 and $x = 4$ or $x = 9$

(b) Let $u = 2^x$: $2u^2 - 9u + 4 = 0 \Rightarrow (2u - 1)(u - 4) = 0$, so $2^x = \frac{1}{2}$ or $2^x = 4$. Hence $x = -1$ or $x = 2$